

Programme	B. Sc. Mathematics Honours			
Course Code	MAT8EJ408			
Course Title	INTRODUCTION TO FRACTALS			
Type of Course	Elective			
Semester	VIII			
Academic Level	400 - 499			
Course Details	Credit	Lecture/Tutorial per week	Practicum per week	Total Hours
	4	4	0	60
Pre-requisites	1. Calculus 2. Geometry			
Course Summary	This course equips students with a thorough understanding of metric spaces and the mathematical foundations of fractal geometry, blending theoretical insights with practical applications.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the basic concepts to build fractals	U	C	Internal Examination/ Assignment/ End Sem examination
CO2	Interpret the dimension of fractals	An	P	Internal Examination/Seminar/ Assignment/ Report/ End Sem examination
CO3	To understand how to construct fractals and apply them	Ap	M	Internal Examination/Seminar/ Report/ End Sem examination
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book	Fractals Everywhere, (2/e), Michael F Barnsley, Dover Publications, 2012			
Module	Unit	Content	Hrs (48+12)	External Marks(70)
I	Metric spaces		15	18
	1	Chapter II, Section 2:- Metric spaces		
	2	Section 3: - Cauchy Sequences, Limit Points, Closed Sets, Perfect Sets, and Complete Metric Spaces		
	3	Section 4: - Compact Sets, Bounded Sets, Open Sets, and Boundaries		
	4	Section 5: - Connected Sets, Disconnected Sets, and Pathwise-Connected Sets		
II	Space of Fractals		15	17
	5	Section 6: - The Metric Space (H(X), h): The Space Where Fractals Live		
	6	Section 7: - The Completeness of the Space of Fractals – up to Theorem 7.1		
	7	Section 7: - The Completeness of the Space of Fractals – From Theorem 7.1 onwards.		
	8	Chapter III, Section 1 – Transformations on the Real line – up to definition 1.3		
	9	Section 1: – Transformations on the Real line – from definition 1.3 onwards.		
	10	Section 2: – Affine Transformations in the Euclidean Plane		
	11	Section 6: – The Contraction Mapping Theorem		
III	Fractal Dimension		8	18
	12: - Section 7: - Contraction Mappings on the Space of Fractals - up to definition 7.1			
	13: - Section 7: – Contraction Mappings on the Space of Fractals – from definition 7.1 onwards			
	14: - Section 8: – Two Algorithms for Computing Fractals from Iterated Function Systems			
	15: - Section 10: – How to Make Fractal Models with the Help of the Collage Theorem.			
	16: - Chapter V, Section 1: – Fractal Dimension – up to Theorem 1.2			
	17: - Chapter V, Section 1: – Fractal Dimension – from Theorem 1.2 onwards.			
IV	Determination of Dimensions		10	17
	18	Section 2: – The Theoretical Determination of the Fractal Dimension – up to Theorem 2.1(including)		
	19	Section 2: – The Theoretical Determination of the Fractal Dimension – rest of the section.		
	20	Section 3: – The Experimental Determination of the Fractal Dimension.		
	21	Section 4: – The Hausdorff-Besicovitch Fractal Dimension – up to and including Theorem 4.2		

	22	Section 4: – The Hausdorff-Besicovitch Fractal Dimension – rest of the section		
V	OPEN ENDED		12	
	Applications of Fractal functions, Fractal interpolation functions, Space filling curves, Construction of Iterated function systems, Applications of Fractals in medical imaging			
References	1. The Fractal Geometry of Nature, Benoît B. Mandelbrot, W.H. Freeman and Company, 1982. 2. Chaos and Fractals: New Frontiers of Science, (2/e), Heinz-Otto Peitgen, Hartmut Jürgens, Dietmar Saupe, Springer, 2004 3. Fractals: Form, Chance, and Dimension, Benoît B. Mandelbrot, W.H. Freeman and Company, 1977. 4. Fractals Everywhere, (2/e), Michael F. Barnsley, Academic Press, 1993. 5. An Introduction to Fractals and Chaos, Michael F. Barnsley, Cambridge University Press, 2021.			

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	2	1	2	1	2	2	2	1	1
CO 2	3	3	1	1	2	1	2	2	2	1	1
CO 3	3	2	2	1	2	1	2	2	2	1	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓

RESEARCH METHODOLOGY